

A LAYERED FORMAL RECONSTRUCTION

Coexistence From First Principles

The kernel and reconstructed structure of Madhyasth Darshan

A checkable map of what depends on what — **three primitives, two axioms, one closure schema**, carried from atoms to societies.

Can a spiritual philosophy carry a logical skeleton?

The texts assert **piecewise** — atoms here, cells there, human society elsewhere. Read that way, the darshan looks like a family of separate teachings.

The claim of this paper: one structure runs underneath all of it. The general case is stated first; the human case is organized as its *most demanding instance*, not the source of the rest.

- **Articulate, don't extrapolate**

State in one structure what the texts assert about atoms, cells, organisms, and human collectives.

- **A lens, not a proof machine**

The formalism clarifies; it proves nothing about the reality of the medium or of jeevan.

- **Keep the seams visible**

Every claim keeps its textual source, its role in the model, and its epistemic standing on show.

Four parts — and one caveat to hold throughout

I

The picture

Before any formalism: the whole thing in four moves, and the discipline that keeps it honest.

II

The kernel

Three primitives, two axioms, one generative move, and the reflexive turn that opens sentence.

III

The structure

Orders, coupling, jeevan and body, justice, time, stability — the reconstruction in working form.

IV

Reach & honesty

The pattern at every tier, comparisons, the formal spine, and where the model stops.

THE CAVEAT

This is not a proof that Madhyasth Darshan is true, nor a reduction of its teaching to one formula. It is a reconstruction — warranted by citations, but the paper's own construction. The formalism organizes an open metaphysical posit; it never establishes it.

PART I

The picture before the formalism

One schema across four orders; the whole move in four steps; a worked example; and the discipline that keeps the reconstruction from overclaiming.

One schema, asserted across the orders

The texts state the universality directly. *“All units saturated in the Omnipresence have form, properties, essential nature & dharma, and inherent orderliness”* (MVD p. 11). The task is therefore **articulation** — one structure for what is asserted piecewise.

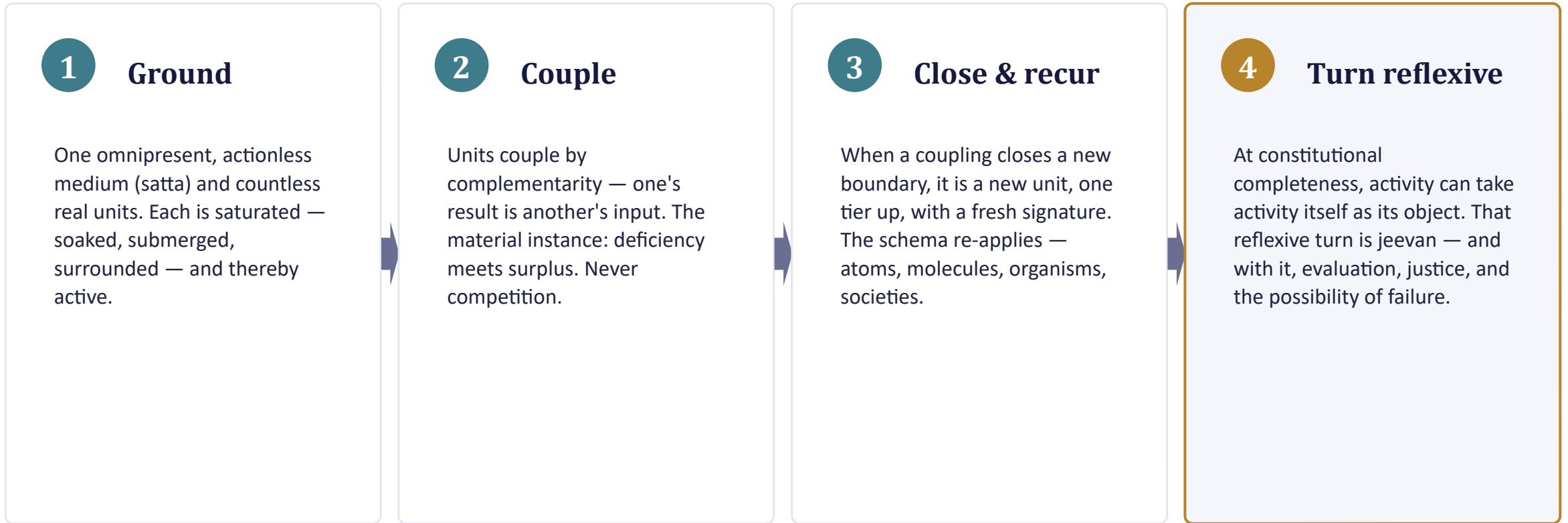
Atoms & molecules	hungry / overfull atoms bond into molecules	MVD p. 8, 42
Cells & organisms	cells compose into living bodies	JV p. 82; SB p. 16
Human collectives	individuals organize into families, undivided society	MVD p. 55, 161

The direction matters

Not *“humans are like atoms.”*

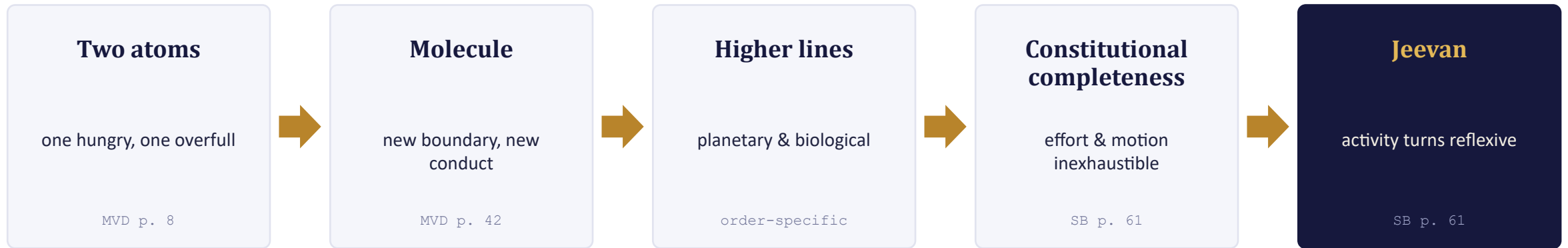
The universal structure is asserted **first**; the human being is its most demanding instance.

Ground, couple, close, and turn reflexive



One closure schema builds the tiers; a distinct reflexive move opens the knowledge order.

A worked example: from two atoms to jeevan



Below the turn — development

Deficiency is met from OUTSIDE: surplus particles supplied by complementary units through closure. Currency: particles-and-bonding.

At & above the turn — realization

Constitution is already complete. Remaining deficiency is met by the jeevan's own activity turning INWARD. Currency: understanding-and-alignment.

Every claim carries a status on two axes

The paper refuses to let elegance stand in for evidence. Each notion is tagged twice — where it comes from, and what the formalism actually establishes.

Textual provenance

- **Textual assertion** — stated in MVD / SB / JV / KD
- **Formal reconstruction** — the paper's own kernel-level organization
- **Mathematical analogy** — the categorical spine, the physical readings

Epistemic standing

- **Established in the model** — a result of the reconstruction
- **Taught datum** — content the texts fix; the model can't produce it
- **Open metaphysical posit** — asserted, not established

The axes cross:

satta and jeevan are textual assertions AND open posits · **k** is a reconstruction established within the model · the six value-kinds are assertions that stay taught data.

PART II

The kernel

Three primitives, two axioms, one generative move — and the reflexive turn where sentence enters. Everything else in the darshan's ontology is organized from here.

Three primitives

O

satta · Omnipresence

One, real, all-pervasive, ACTIONLESS, state-complete, energy- and knowledge-natured (gyan).
Not a unit; no statement over units applies to it.

JV p. 31; MVD p. 12

U

units · ikai

Real, co-eternal, countable CARRIERS. A unit bears a state and an activity — it is never built from activity or process.

JV p. 45

—

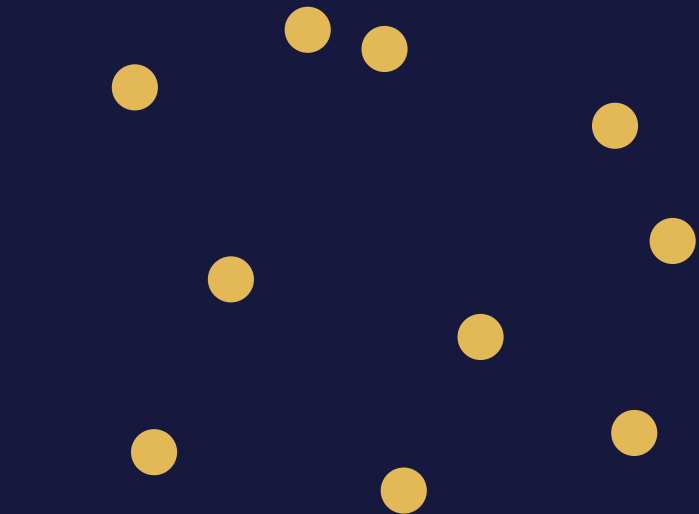
saturation

The one ground relation: every unit soaked, submerged, surrounded in the single O.
Constitutive co-location, not a drawing of energy from O.

JV p. 18; SB p. 49

O — satta

the medium (actionless)



units (ikai) saturated in the one medium

Activeness is NOT a primitive. “Unit + Energy fullness = Activeness” (SB p. 69) — a co-presence conferred by saturation, recovered as a theorem, not a fourth building block.

Two axioms — endowment and orientation

A1 Endowment — the satta firewall

Through saturation, O endows each carrier with three inseparable invariants:

- (a) **Activeness** the triad — effort (shram), motion (gati), result (parinam)
- (b) **Recognition** activity is unit-to-unit legible, by law
- (c) **Regulation** lawfulness disciplines a unit's natural-state conduct

THE FIREWALL

O is never an input to, nor a summand of, a unit's activity. It provisions energy, legibility, and regulation — with no push, no update term, no depletion.

A2 Completeness-orientation

Every unit's activity is directed toward **development and completeness** — not toward indefinite quantitative increase. *“each unit... is active and developing solely for completeness” (SB p. 51).*

Terminal form is TIERED. The atomic line culminates at **constitutional completeness** — the threshold where jeevan appears (T1). The realized goals — restfulness of effort, destination of motion, immortality of result — are the awakening targets of jeevan.

κ — compound-closure

Given saturated units whose ports are **complementary**, their activities couple; when the coupling closes a new motion-path, they present one bounded whole — a new unit, one tier up, with a fresh signature.

$u_1, \dots, u_n \in U_o$ saturated in O $C_o(u_1, \dots, u_n)$ closure
of a new motion-path

$\kappa(u_1, \dots, u_n) \in U$ — a new unit, new signature, +1 tier

$\kappa_o(u_1, \dots, u_n) = \text{closure}(\phi_o(\rho_o(u_1, \dots, u_n)))$

The anatomy of a coupling

ρ **recognition** identifies C_o holds; fixes the expectation profile

ϕ **fulfilment** executes; results flow along that profile

closure does it settle into one new bounded path?

Compound (yaugik) — closes

A new boundary forms. Components “discard their own conducts and present another kind of conduct.” A new unit exists.

Mixture (mishran) — does not

ϕ still delivers results, but no new boundary. Components keep their conducts. This is why “composition is not development.”

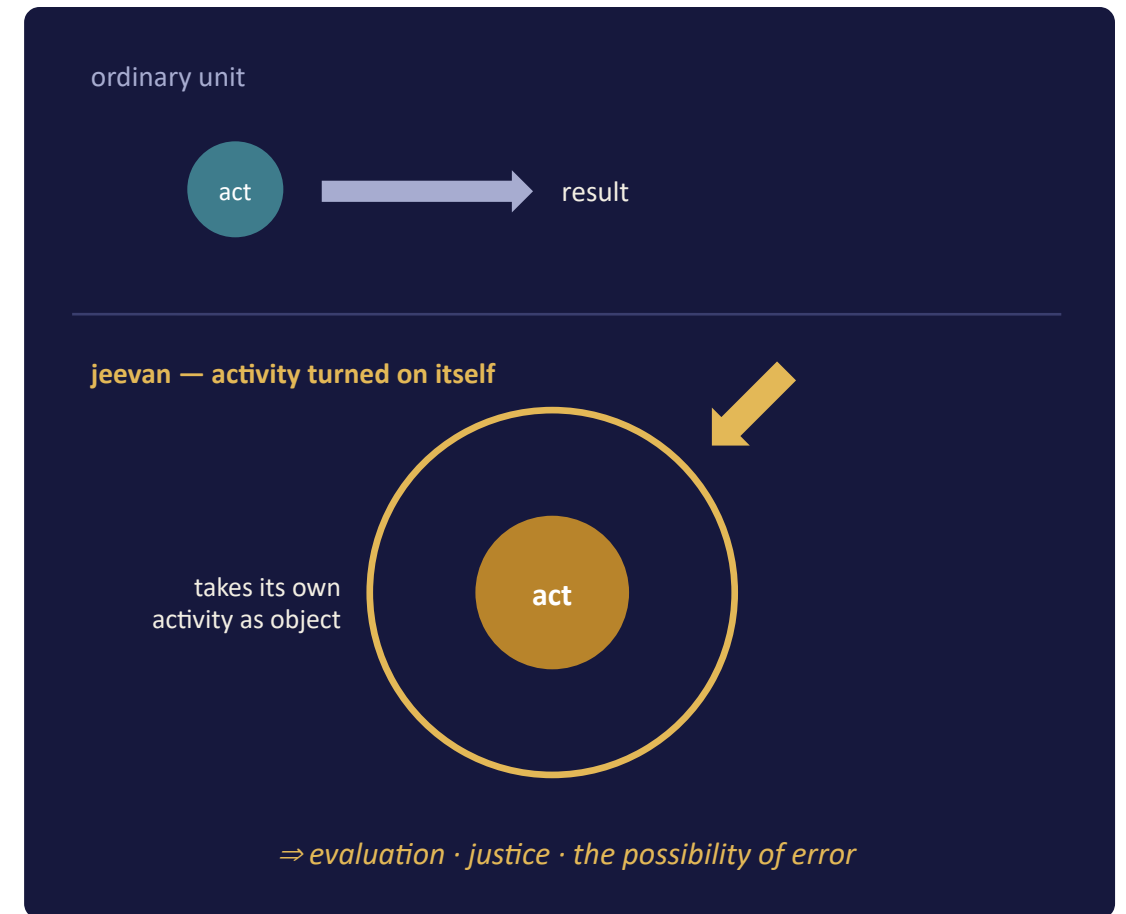
The reflexive turn — κ reaches a fixed point

Iterate κ along the development progression (vikas-kram) until a composite atom reaches its A2 terminal — **constitutional completeness**, after which “both effort and motion become inexhaustible” (SB p. 61).

At that fixed point the unit's activity can take **activity itself** as its object. This reflexive re-application is **jeevan** — the sentient unit, and the sole locus of evaluation and of possible failure.

Status flag

Reflexivity is a DISTINCT move — not a second literal firing of κ . It is the paper's interpretive MODEL of the knowledge order — an organizing reading, flagged as such, not a derived theorem.



PART III

The reconstructed structure

Orders and the regulation ladder, the anatomy of coupling, jeevan and its body, justice and the human goals, time, and stability — the working body of the model.

Category theory, in just enough words

Category theory swaps “*what is each thing made of?*” for “*how does each thing relate to everything else?*” Three ideas carry the whole spine — the rest is optional.



Algebra

build UP

A thing built up by a rule — the home of “generated.” Units of nature = the algebra generated by κ , so definitions become recursion on how a unit was built.



Coalgebra

unfold DOWN

A carrier plus a map that unfolds its behaviour — WITHOUT identifying the carrier with that behaviour. Keeps a unit a substance that bears activity, not a bundle of relations.



Colimit

glue SIDEWAYS

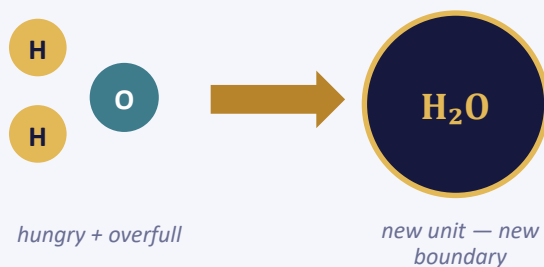
Joins pieces into one whole along what they share — the shape of families joining into one society.

Objects are things to talk about; arrows (morphisms) are relationships; composition chains them. That is all you need for the spine.

Algebra in action — compound and mixture

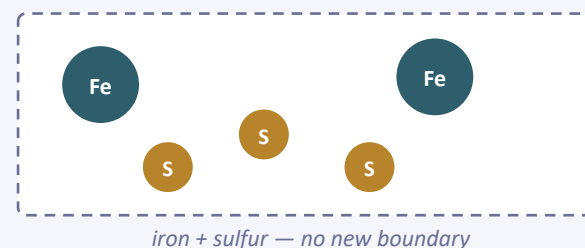
Algebra means **built up by a rule**. The rule is κ . “Units of nature are the algebra generated by κ ” — so to know a unit, you ask *how it was assembled*. Two atoms show the two outcomes.

Compound (yaugik) — closure SUCCEEDS



Deficiency meets surplus (MVD p. 8); the coupling closes a new motion-path. Components “*discard their own conducts, and present another*” (MVD p. 42). A fresh signature is minted — water’s roop, gun, svabhav, dharma. One rung of *vikas*.

Mixture (mishran) — closure FAILS



a magnet still lifts the iron out

Fulfilment (ϕ) still delivers a result — they interpenetrate — but nothing closes. Components “*maintain their respective conducts*” (MVD p. 42); the mixture is separable, no new unit. Hence “**composition is not development**” (SB pp. 75–76).

The algebra, for a practitioner: every real unit carries its build-history, and its properties are read from how it was assembled. A mixture is typed OFF that inductive sort — a mere aggregate. **Vikas happens only by closure, never by accumulation.**

Coalgebra in action — a unit unfolds, yet is never reduced to it

Coalgebra means a **carrier** plus a map that **unfolds its behaviour one step at a time** — without ever saying the carrier IS that behaviour. The darshan's *ikai* is exactly such a carrier: it bears activity; it is never built out of activity.

behaviour map $x \rightarrow F(x)$: read one step of conduct, get the next state



each step emits conduct — the triad (effort → motion → result) and its couplings; in jeevan, via projection–reflection (paravartan–pratyavartan) through atma · buddhi · chitta · vritti · mun

... the unfolding never terminates in a “final” object

The carrier exceeds its behaviour

- **The model refuses the “final coalgebra.”**

No amount of observed conduct is ever allowed to BECOME the unit. The carrier stays prior to, and larger than, its behaviour-stream.

- **So jeevan stays a real ikai.**

A substantial gathanpurna parmanu, co-eternal — not the sum of its aachran, not a function of the body, not a role.

- **Behaviourism inverted.**

“You are what you do” collapses carrier into behaviour. MD (and this reading) says: you HAVE conduct; you are not identical to it.

Two directions, one substance: algebra builds the unit UP (how it was composed) · coalgebra lets it live its conduct DOWNWARD (how it behaves) — and **neither reduces the ikai**. A1 grounds that the behaviour exists and is lawful; O stays actionless, contributing no term.

Four orders — one law, four conformance regimes

Each unit carries a **signature** — form (roop), properties (gun), essential nature (svabhav), dharma — minted fresh at each tier. The four orders differ in *how a unit's method survives*.

ORDER	ADDED CAPABILITY	REGULATION REGIME	MODE
<i>Material</i>	recognising, fulfilling	structural conformance	definite
<i>Pranic / bio</i>	+ selection	seed conformance (beej)	definite
<i>Animal</i>	+ sensation, hope	species conformance (vansh)	definite
<i>Knowledge</i>	+ knowing, believing, evaluation, choice	sanskar conformance	achieved

The first three orders are DEFINITE and inerrant — regulated by constitution. **Only the knowledge order is “achieved” — routed through a self-model that can misfire. That is why the human tier is the one unfinished tier.**

Recognition, fulfilment, and the gate of capacity



The closed knowledge-order cycle $p \rightarrow \varphi \rightarrow \mu \rightarrow$ mutual satisfaction is what the texts call justice.

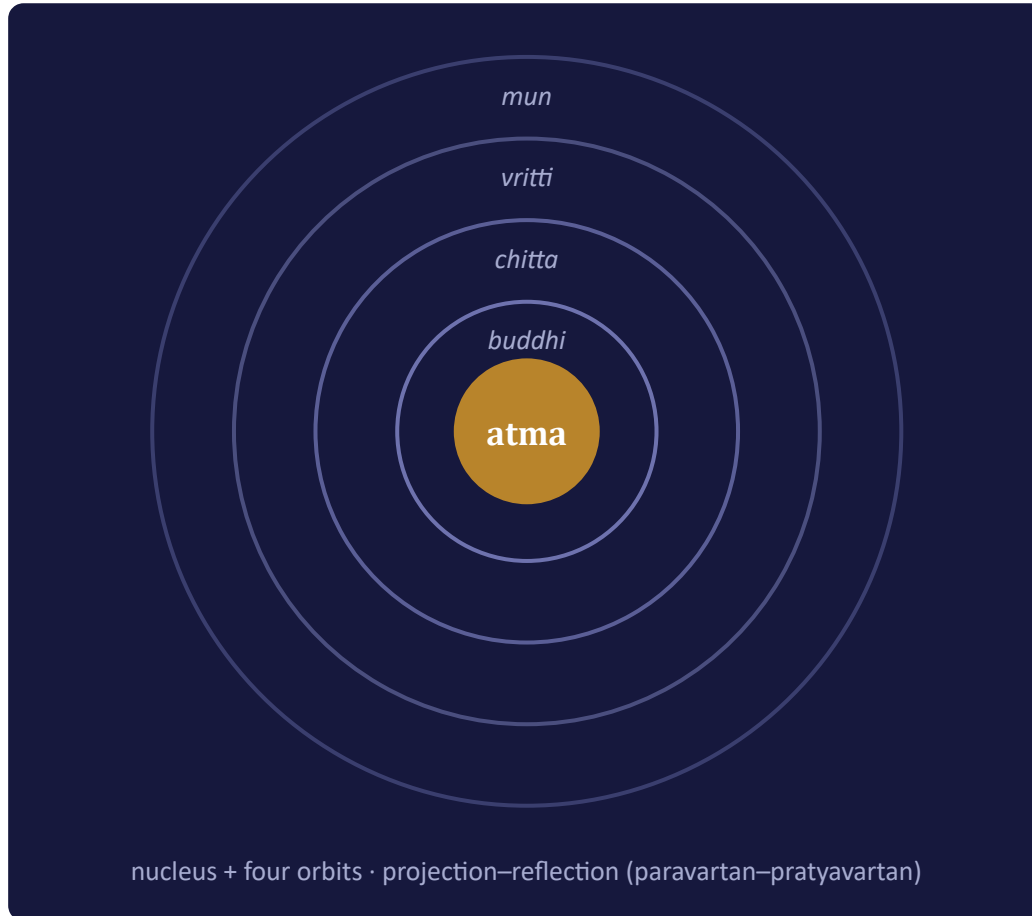
Fulfilment is gated by capacity

capacity (kshamata) · ability (yogyata) · receptivity (patrata)
 φ is realised at the level these permit — not by receptivity alone.

The link back to κ

Recognition and fulfilment are the two phases of EVERY coupling. **The generative move κ is simply what a fulfilled coupling BECOMES when it closes.**

Jeevan and its body — nucleus, orbits, expression-gate



atma authenticity, realisation · **buddhi** resolve, enlightenment

chitta visualisation, contemplation · **vritti** analysis, deliberation

mun selection, taste

Faculty harmonies (H) — non-scalar

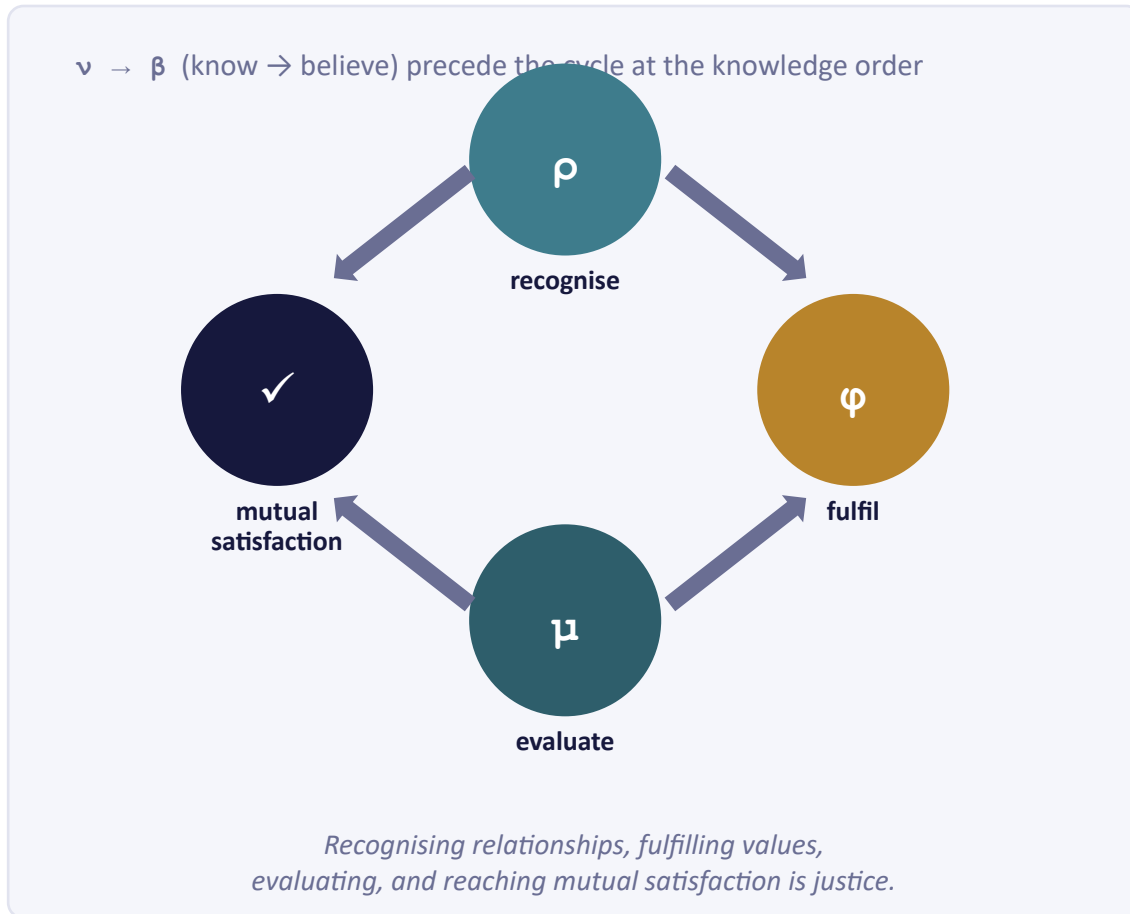
mun–vritti → happiness vritti–chitta → peace

chitta–buddhi → contentment buddhi–atma → bliss

The body is an expression-gate

jeevan's capability is intrinsic and complete; how much can FUNCTION is fixed by the body. The human tier is gated **twice** — by an understanding that can fail, and by a body that must be human (the fifth kosha, vighnamaya, opens only in a developed medhas-body).

Justice — the closed cycle — and the four goals



Justice at scale evidences four goals

- 1 **Resolution** — in the person
- 2 **Prosperity** — in the family
- 3 **Fearlessness** — in society
- 4 **Coexistence** — in universal order

Provisions of coexistence — not automatic outcomes.

Time is the duration of activity — never its cause

“Duration of an activity is referred to as time” (MVD p. 34). Time enters not as a background substance or cause, but as a **measure of activity**.

A duration grading δ on unit-executions

$$\delta(\text{id}_u) = 0 \qquad \delta(e_2 \circ e_1) = \delta(e_1) \oplus \delta(e_2)$$

The identity is a UNIT's no-change execution — not satta. Since O is not a unit and performs no activity, **$\delta(O)$ is undefined**, not zero.

Kriya-shunya $\Rightarrow$ kaal-shunya

Where there is no activity, duration has no application — not because O is a zero-duration event. O's presence “at all times” names its omnipresence with every active unit, not becoming within O.

Time does not:

select a path

a progression orders states toward A2; time only records how long

supply energy

that is A1's endowment through saturation

orient toward completeness

that is A2's orientation

The natural state as attractor — a guarded analogy

A unit is in its **natural state** when its couplings are fulfilled in conducive conditions, and **excited** otherwise — “each unit moves towards development in its natural state and decline in its excited state.”

A conditional Lyapunov analogy

$$K_o(x) = 0 \quad \text{iff} \quad x \in N_o$$

$$K_o(\Phi_o(t, x)) \leq K_o(x) \quad \text{during unforced relaxation}$$

K_o measures deviation from the natural set N_o . N_o is an active regime, not inactivity — activeness continues when $K_o = 0$.

Local and conditional — not a law

- The inequality is local — perturbation, adverse environment, or error may raise K_o .
- No global monotonicity theorem is claimed.
- T1 (the passage to sentience) is a guarded, irreversible threshold — not an asymptote every trajectory approaches.

What makes it the darshan's — not physics'

Minimizing K_o is bounded, constrained optimization — but the TARGET is a qualitative natural set, not indefinite increase. *Hamilton's principle is a contrast, not the governing model*; and at the knowledge order no single scalar suffices — fulfilment, justice, harmony, and participation form a preorder, not one maximand.

PART IV

Reach, comparison, and honesty

The same column-shape at every tier; what the structure is NOT; where it sits among neighbouring frameworks; the formal spine; and a frank accounting of its limits.

The same column-shape, four times

The structure is universal because the column repeats — **asserted for each order, not carried by metaphor from human life.** *(Abridged; the paper's table has thirteen rows.)*

	Material	Biological	Animal	Knowledge
Regulation	structural	seed	species	sanskar; justice
Value	utility + art	utility in food-chains	+ sensitivity	+ evaluation (μ)
Recog.–fulfil	definite	definite	definite	achieved
Assembly	literal compound	compositional	herd / lineage	conduct-compound (analogy)
Transmission	constitution	seed (rachna vidhi)	lineage	education-sanskar
Failure mode	—	death / decompose	species stress	delusion → war, overdraw

Every row is richer in the knowledge order but never ABSENT below it. **What first appears with humans is not value — it is the EVALUATION of value, and with it, error.**

What the structure is NOT

The most natural modern readings quietly swap the structure for something else. Four corrections:



Not driven by indefinite increase

Conduct in the first three orders is definite; object-usefulness is constant; “more” is the diagnosed PATHOLOGY, not the telos. Completeness is a qualitative terminal condition.



Not struggle or selection

The assembly engine is complementarity, explicitly opposed to conflict and survival-of-the-fittest. Decomposition is the LOSS of structure, not its designer.



Not bottom-up emergence alone

Capacities are provisioned in existence and manifest when composition reaches the right tier; O energises every tier at once. Drop O and energisation becomes magic.



Not value-as-epiphenomenon

Value is the content of fulfilment at EVERY tier; the human innovation is a second-order operation on it, located in jeevan — “not bodily or mechanical.”

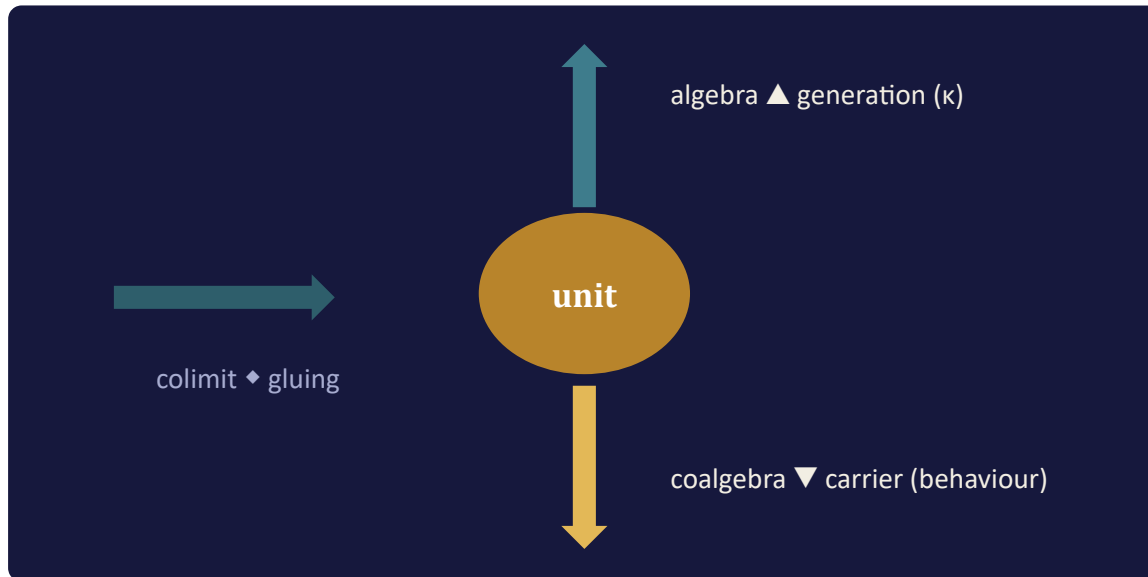
Neighbours — and two things none of them have

FRAMEWORK	SHARES	DIFFERS
Holarchy (Koestler)	every unit is whole and part	no medium O; no value in relations; no transmission
Autopoiesis (Maturana & Varela)	persistence by self-production	living systems only; here: atoms→societies, persistence = fulfilment
Multilevel selection	nested units persist or fail as wholes	selection is competitive & stochastic; here: complementarity & definite
Thermodynamic self-org.	structure from flows between units	needs free-energy throughput; here units are energised constitutively
		silent on WHICH compositions occur: complementarity is the missing

1 The medium O — constitutive, energising, regulating at every tier at once. A precondition OF physics, not an aether within it.

2 Value everywhere — the universal content of relations, with EVALUATION appearing only at the top tier.

The formal spine — and the telos it drives toward



Downward keeps units SUBSTANTIAL — **the carrier is not quotiented by behaviour; the theory refuses the final coalgebra**. This is what keeps jeevan a real entity, not a functional role — while O stays actionless.

Sideways encodes **undivided society (akhand samaj)**: families glue along shared members — coherent exactly when local value-structures AGREE on what they share (a descent condition).

Undivided society is the telos — not an incidental output

The four human goals are the shared value-content every jeevan aspires to; the colimit glues coherently precisely when that content is recognised across members — **compatible in the awakened case, degenerate in delusion** (“united in the right, divided in the wrong”, JV p. 17, 40).

Limitations, heavy posits, and open problems

● Formalisation is interpretive

The symbols (O , κ , ρ , ϕ , μ , δ ...) are the paper's compression, not in the texts.

● The residual premise

Universal coupling rests on no-isolation (JV p. 43) — adopted, not derived. The one place the kernel leans beyond its axioms.

● Translation risk

KD is a machine-assisted WORKING translation — corroboration only, to be checked against the Hindi before release.

● No empirical contact

Nothing here yields a measurement or prediction. “Elegant diagrams can create an illusion of proof.”

The heavy posits, named

- A2's teleology (orientation to completeness)
- The mediative regulation asserted in A1
- Selection predicate: which configuration reaches T1
- Selection predicate: which jeevan animates which body

A sample of the open problems

Countermodels for the independence of A1 and A2 · discharge or confirm no-isolation · identify the order-relative relations C_o · a mechanised recovery audit · concurrency in the duration model · formalize the non-scalar preorders H and S · physical variables for K_o and testable dynamics.

CONCLUSION

The ground is coexistence itself

One actionless medium and countless real units, bound by saturation. A1 provisions activeness, recognition, and regulation; A2 supplies orientation toward completeness; typed complementarity and closure organize matter, biology, and the declared analogy of humane social order.

● One schema, four orders

Articulated, not extrapolated — the human case is the hardest instance of a universal structure.

● Sentience is a reflexive turn

At constitutional completeness activity takes itself as object — the sole locus of evaluation, and of error.

● Honesty by construction

Provenance × standing keeps analogy from masquerading as proof — a map of what depends on what.

Strongest when every claim keeps its provenance, role, and standing visible — especially where a mathematical analogy organizes an open posit without establishing it.

DISCUSSION

Questions & discussion

Where would you like to push? — the reality of the medium, the teleology of A2, the sentience threshold, the society-gluing condition, or testability.